Practice Exam - 1

Question 97 of 117

A 72-year-old woman presents to the outpatient clinic with severe, gradually worsening exertional dyspnea. Symptoms include shortness of breath with one flight of stairs and light household chores. Upon examination, BMI is 23 kg/m2, heart rate is 65 bpm, jugular venous pressure is 8 cm of water, and there is no peripheral edema. Electrocardiogram shows sinus rhythm and normal intervals. A pharmacologic nuclear stress test is negative for infarction and inducible ischemia; coronary artery calcium score was 0.

Echocardiogram reveals a left ventricular ejection fraction of 65%, LV wall thickness of 1.0 cm throughout, mitral calcification with mild mitral regurgitation and moderate mitral stenosis with a diastolic mitral valve gradient of 6 mm Hg (at a heart rate of 65 bpm) and mitral valve area of 1.5 cm2. Right ventricular size and function are normal, with tricuspid regurgitation jet velocity of 2.0 m/s.

What is the next best diagnostic test?

A. Coronary angiography

B. Exercise right heart catheterization

C. Exercise positron emission tomography stress test

D. Exercise echocardiography

Commentary | References

Testing Point: The Board-certified AHFTC specialist should be able to appropriately evaluate valvular heart disease in heart failure patients.

Answer: D. Exercise echocardiography.

According to the ACC/AHA Guidelines, provocative exercise testing is warranted to evaluate any valvular disease where there is a significant discrepancy between symptoms and valve disease severity based on resting assessment. In this case, although the echocardiogram report concludes moderate mitral stenosis, the gradient and mitral valve area are actually borderline for severe mitral stenosis. The patient has demographics and echo features consistent with rheumatic heart disease too. Exercise echo would reveal pathologic elevation in mitral valve gradients and estimated RVSP. Answer B: An exercise right heart catheterization may provide similar information about valvular disease, but a shunt series would not provide any information of utility in this case.

Normal electrocardiogram and lack of paroxysmal symptoms argues against arrhythmic causes. Normal right heart features on echocardiogram argues against pulmonary hypertension.

Answer A: Reassuring ischemic evaluation argues against coronary artery disease.

Answer C: PET stress testing would not add much to the already performed and reassuring pharmacologic nuclear stress test.